

Section 13.1.

1. Show that $p(x) = x^3 + 9x + 6$ is irreducible in $\mathbb{Q}[x]$.

And, for a root θ of $p(x)$, find $(1+\theta)^{-1}$ in $\mathbb{Q}(\theta)$.

Proof. Since $9, 6 \in (3)$ but $6 \notin (3^2)$, so the Eisenstein Criterion gives p is irreducible.

Let θ be the root of $p(x)$. By the Division Algorithm,

$$x^3 + 9x + 6 = (x+1)(x^2 - x + 10) - 4$$

$$\text{Thus, } 4 = (\theta+1)(\theta^2 - \theta + 10) - (\theta^3 + 9\theta + 6) = (\theta+1)(\theta^2 - \theta + 10), \text{ or}$$

$$(\theta+1)^{-1} = \frac{1}{4} \cdot (\theta^2 - \theta + 10).$$

2. Show that $x^3 - 2x - 2$ is irreducible over $\mathbb{Q}$ and let θ be a root.

Compute $(1+\theta)(1+\theta+\theta^2)$ and $(1+\theta)(1+\theta+\theta^2)^{-1}$.

$$\text{Proof. } (1+\theta)(1+\theta+\theta^2) = 1 + \theta + \theta^2 + \theta + \theta^2 + \theta^3 = \theta^3 + 2\theta^2 + 2\theta + 1 \\ = 2\theta^2 + 2\theta + 1 + 2\theta + 2 = 2\theta^2 + 4\theta + 3.$$

And,

$$(x^3 - 2x - 2) = (x^2 + x + 1)(x - 1) + (-2x - 1)$$

$$(x^2 + x + 1) = (-2x - 1)\left(-\frac{1}{2}x - \frac{1}{4}\right) + \frac{3}{4}$$

$$\Rightarrow (x^2 + x + 1) = (x^3 - 2x - 2)\left(-\frac{1}{2}x - \frac{1}{4}\right) - (x^2 + x + 1)(x - 1)\left(-\frac{1}{2}x - \frac{1}{4}\right) + \frac{3}{4}$$

$$\text{So, } (\theta^2 + \theta + 1) = 0 - (\theta^2 + \theta + 1)(\theta - 1)\left(-\frac{1}{2}\theta - \frac{1}{4}\right) + \frac{3}{4},$$

$$(\theta^2 + \theta + 1)\left(-\frac{1}{2}\theta^2 + \frac{1}{4}\theta + \frac{5}{4}\right) = \frac{3}{4},$$

$$\text{So, } (\theta^2 + \theta + 1) = \frac{4}{3}\left(-\frac{1}{2}\theta^2 + \frac{1}{4}\theta + \frac{5}{4}\right)$$

$$= \frac{1}{3}(-2\theta^2 + \theta + 5).$$

#13.1.3. Show that x^3+x+1 is irreducible over $\mathbb{F}_2$.

And let θ be a root of the polynomial. Compute θ^n in $\mathbb{F}_2(\theta)$.

Proof. $0+0+1 \neq 0$, $1+1+1 \neq 0$, So, x^3+x+1 has no root in $\mathbb{F}_2$,
and it has degree 3, So irreducible.

And, $\theta = \theta$, $\theta^2 = \theta^2$, $\theta^3 = -\theta - 1 = \theta + 1$ (in $\mathbb{F}_2$)

$$\theta^4 = \theta^2 + \theta$$

$$\theta^5 = \theta^3 + \theta^2 = \theta^2 + \theta + 1$$

$$\theta^6 = \theta^3 + \theta^2 + \theta = \theta^2 + 2\theta + 1 = \theta^2 + 1$$

$$\theta^7 = \theta^3 + \theta = 2\theta + 1 = 1.$$

Thus, $(\mathbb{F}_2[x])/(x^3+x+1)$ is a field, order of 8.

#13.1.5. Let α is a rational root of a monic polynomial in $\mathbb{Z}[x]$.

Prove that α is integer.

Proof. Let $\alpha = r/s$, where r and s are relatively prime integers. Then,

$$\frac{r^n}{s^n} + a_{n-1} \cdot \frac{r^{n-1}}{s^{n-1}} + \dots + a_1 \cdot \frac{r}{s} + a_0 = 0.$$

$$\Rightarrow r^n + a_{n-1} \cdot s r^{n-1} + \dots + a_1 s^{n-1} r + a_0 s^n = 0.$$

$$\Rightarrow r^n = s \cdot (-a_{n-1} r^{n-1} - a_{n-2} s r^{n-2} - \dots - a_0 s^{n-1})$$

That is, s divides r^n . But, since $(s, r) = 1$, it is contradiction.

#13.1.7. Prove that x^3-nx+2 is irreducible, except $n = -1$ or 3 or 5 .

Proof.

Sec 13.2.

3. $x = 1+i$. $(x-1)^2 = -1$, so $x^2 - 2x + 2 = 0$.

It is minimal, because $1+i \notin \mathbb{Q}$.

4. Determine the degree over $\mathbb{Q}$ of $2+\sqrt{3}$ and $1+\sqrt[3]{2}+\sqrt[3]{4}$.

Proof. $2+\sqrt{3} \in \mathbb{Q}(\sqrt{3})$, so $\mathbb{Q}(2+\sqrt{3}) \subseteq \mathbb{Q}(\sqrt{3})$, and since $\mathbb{Q} \neq \mathbb{Q}(2+\sqrt{3})$, and $[\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = 2$, so $\deg 2+\sqrt{3} = 2$.

And, since $1+\sqrt[3]{2}+\sqrt[3]{4} \in \mathbb{Q}(\sqrt[3]{4})$, again, similarly, $\deg 1+\sqrt[3]{2}+\sqrt[3]{4} = 3$.

5. Let $F = \mathbb{Q}(i)$. Prove x^3-2 and x^3-3 are irreducible over F .

Proof. Since both are degree 3, the one is irreducible iff one has a root.

Let α be a root in $\mathbb{Q}(i)$ of x^3-2 . Then, $\mathbb{Q}(\alpha)$ has degree 3,

since x^3-2 is irreducible over $\mathbb{Q}$, by the Eisenstein criterion.

But, $[\mathbb{Q}(i) : \mathbb{Q}] = 2$, this is contradiction to $\mathbb{Q}(\alpha) \subseteq \mathbb{Q}(i)$.

Similarly x^3-3 is irreducible.

7. Prove that $\mathbb{Q}(\sqrt{2}+\sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

Proof. Clearly, $\sqrt{2}+\sqrt{3} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$, so $\subseteq$ is proved.

Conversely, $\sqrt{2}+\sqrt{3} = x \Rightarrow 2 = x^2 - 2\sqrt{3}x + 3$

$$\Rightarrow 2\sqrt{3}x = x^2 + 1$$

$$\Rightarrow 12x^2 = x^4 + 2x^2 + 1$$

$$\Rightarrow x^4 - 10x^2 + 1 = 0.$$

It is irreducible, because $(x^2)^2 - 10(x^2) + 1 = 0$ has no root in $\mathbb{Q}$.

So, $\deg_{\mathbb{Q}} \sqrt{2}+\sqrt{3} = 4$. Meanwhile, since $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$, so these are same.

#8, 9. Biquadratic.

Let F be a field of characteristic $\neq 2$.

Let $D_1, D_2 \in F$, neither of which is a square in F .

Prove that: If $D_1 D_2$ is not a square in F , then $F(\sqrt{D_1}, \sqrt{D_2})$ is degree 4.

If $D_1 D_2$ is a square in F , then $F(\sqrt{D_1}, \sqrt{D_2})$ is degree 2.

Proof. First, $[F(\sqrt{D_1}) : F] = 2$ is clear.

So, enough to show that $[F(\sqrt{D_1}, \sqrt{D_2}) : F(\sqrt{D_1})] = 2$

If these are same, then $\sqrt{D_2} \in F(\sqrt{D_1})$, so $\sqrt{D_2} = a_1 \sqrt{D_1} + a_2$ for some $a_1, a_2 \in F$.

Now, $D_2 - a_2^2 D_1 = a_2(\sqrt{D_2} + a_1 \sqrt{D_1}) = a_2(2a_1 \sqrt{D_1} + a_2) = 2a_1 a_2 \sqrt{D_1} + a_2^2$.

$\Rightarrow 2a_1 a_2 \sqrt{D_1} = D_2 - a_2^2 D_1 + a_2^2$.

$2 \neq 0$, since $\text{ch}(F) \neq 2$, and $a_1 \neq 0$, because $a_1 = 0$, then $\sqrt{D_2} = a_2 \in F$, and $a_2 \neq 0$; if $a_2 = 0$, then $\sqrt{D_2} = a_1 \sqrt{D_1} \Rightarrow \sqrt{D_2} = a_1 \in F$.

So, $\sqrt{D_1} = (2a_1 a_2)^{-1}(D_2 - a_2^2 D_1 + a_2^2) \in F$, contradiction.

If $\sqrt{D_1 D_2} \in F$, then $\sqrt{D_1} \in F(\sqrt{D_2})$, so prove.

Let $a, b \in F$ and b is not a square in F . Then,

$\sqrt{a+b} = \sqrt{m} + \sqrt{n}$ for some $m, n \in F$ iff $\sqrt{a^2-b} \in F$.

Proof. Observe that if $\sqrt{a+b} = \sqrt{m} + \sqrt{n}$ for some $m, n \in F$, then

$\sqrt{a+b} = \sqrt{m} + \sqrt{n} \Rightarrow a+b = m+n+2\sqrt{mn} \Rightarrow a-(m+n) = 2\sqrt{mn}-\sqrt{b} \in F$,

Then $2\sqrt{mn}-\sqrt{b}$ must be zero; let $2\sqrt{mn}-\sqrt{b} = k \in F$. Then

$4mn = k^2 + b + 2k\sqrt{b}$. If $k \neq 0$, then $\sqrt{b} = \frac{1}{2k} \cdot (k^2 - b + 4mn)$, so $\sqrt{b} \in F$.

(And $2 \neq 0$, by $\text{ch}(F) \neq 2$)

So $2\sqrt{mn} = \sqrt{b}$ and $a = m+n$, so $a^2 - b = (m+n)^2$.

Conversely, $m = \frac{1}{2}(a + \sqrt{a^2-b})$ and $n = \frac{1}{2}(a - \sqrt{a^2-b})$, then we obtained the left.

Now, $\mathbb{Q}(\sqrt{a+b})$ ($a, b \in \mathbb{Q}$) has degree 4 iff $a^2 - b$ is square in $\mathbb{Q}$.

So, $\mathbb{Q}(\sqrt{3+2\sqrt{2}})$ has degree 4. $9-8=1$.